

Learning to Sparsify Differences of Synaptic Signal for Efficient Event Processing

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Abstract

Neural network processors on edge devices need to process spatiotemporal data with low latency, which requires a large amount of multiply-accumulate operation (MAC). In this paper, we propose a difference-driven neural network framework for efficient video or event stream processing. Our framework achieves lower MAC by learning to sparsify the *temporal differences of synaptic signals* (TDSS) of proposed masked convolutional neural networks. By reducing the TDSS, MAC reduction is achieved in a unified manner by increasing the quantization step size, disconnecting synapses, and learning weights that respond sparsely to inputs. A novel quantizer is another key to realize unified optimization; the quantizer has a gradient called *macro-grad* that guides the step size to reduce the MAC by reducing the TDSS loss. Experiments conducted using a wide range of tasks and data (frames/events) show that the proposed framework can reduce MAC by a factor of 32 to 240 compared to dense convolution while maintaining comparable accuracy, which is several times better than the current state-of-the-art methods.

1 Introduction

Edge devices such as autonomous vehicles and mobile phones often use neural networks (NNs) to process spatiotemporal data such as videos. Processing such data independently in a frame-by-frame manner requires intensive computation.

Approaches for the efficient processing of spatiotemporal data from both algorithmic and processor perspectives are explored [82, 44, 46, 48]. Data from temporal proximity are highly correlated; that is, *differences* between each pixel from consecutive frames are highly sparse. The $\Sigma\Delta$ network [46] was proposed for efficient video processing; this is a reformulation of quantized neural networks into equivalent quantized recurrent neural networks. The network recursively updates its state and output using the sparse temporal difference. The sparse update mechanism results in lower multiply-accumulate operation (MAC). The model has been

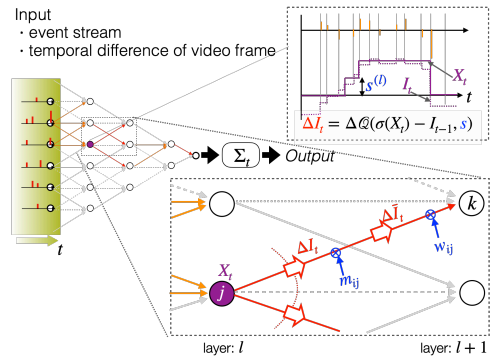


Figure 1: **TDSS-aware training.** In an $m\Sigma\Delta$ network, only the masked difference $\Delta\bar{l}$ is processed. MAC could be reduced jointly by weight w , mask m , and quantization step size s . We realize the optimization by the novel quantizer and loss function.